

PINCHEN XIE

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POSITION

Luis W. Alvarez Postdoctoral Fellow, Lawrence Berkeley National Laboratory 2024-2027

EDUCATION

Ph.D. Applied and Computational Mathematics, Princeton University 2018-2024
Advisor: Roberto Car (Chemistry, Physics) and Weinan E (Math)
B.S. Physics, Fudan University 2014-2018

AWARDS

Young Researcher Award, International Workshop on Relaxor Ferroelectrics 2026
Developers of the Year, Lawrence Berkeley National Laboratory 2026
UC President's Lindau Nobel Laureate Meeting Fellowship, University of California 2025
Luis W. Alvarez Fellowship, Lawrence Berkeley National Laboratory 2024-2026
Chun-Tsung Scholar, H. C. Chin and T. D. Lee Undergraduate Research Endowment 2017

RESEARCH INTERESTS

Methodology and application of machine learning-assisted, *ab initio* cross-scale ($\text{\AA} \sim \mu\text{m}$) modeling of solid-state functional materials, biomolecules, and open quantum systems.

GRANTS

Co-PI (with Erika Ye and Chao Yang), IBM Quantum Innovation Center. 2026. Lindblad simulation of Stochastic Differential Equations via Universal Dilation
Lead PI (with Anupam Mitra), LDRD, LBNL. 2026-2027. Predictive, Hardware-Adaptive Quantum Error Correction.
Lead PI LDRD, LBNL. 2024-2026. ML-assisted *ab initio* multi-scale modeling.
Lead PI AI4Sci@NERSC, DOE. 2025-2026. ML-assisted coarse-graining for routine simulation of billion-atom ferroic devices

SERVICE

Referee for
• npj Comput. Mater. • J. Chem. Theory Comput. • J. Phys. Chem. • Quantum Mach. Intell. • J. Mater. Chem. A • New J. Phys. • ICLR
Conference Service
Session Chair, MRS Spring Meeting, Hawaii 2026
Program Committee, IEEE International Parallel & Distributed Processing Symposium (IPDPS) 2026
Workshop and Seminar Organization
Tutor, Deep Modeling for Molecular Simulation Workshop, Princeton 2022-2024

TEACHING AND ADVISING

Intern Student Mentor - Berkeley Lab
Xinyu Xu (M.S., Stanford); Ruiqi Gao (Ph.D., Princeton); Siyi Huang (B.S., UC Berkeley) 2025
Xinyu Xu (Ph.D., Princeton); Ruiqi Gao (Ph.D., Princeton); Jia Wang (Ph.D., UIUC); 2026

Assistant Instructor - Princeton University

PHY512 Monte Carlo and Molecular Dynamics Simulation

MAT202 Linear Algebra with Applications

MAT199 Math Alive

MAT201 Multivariable Calculus

Spring 2022 & 2023

Fall 2021

Spring 2020

Fall 2019

OPEN SOURCE SOFTWARE

Lead contributor, PeroStruc, package for sampling compositional and geometric structures of complex perovskites

Lead contributor, QEpsilon, GPU-enabled toolkit for dynamical modeling of open quantum systems

Lead contributor, OpenFerro, unified framework for lattice simulation with billion-atom capability on GPUs

Lead contributor, AIGLETools, toolkit for non-Markovian molecular dynamics in OPENMM

PUBLICATIONS

1. X. Xu, K. Cai, Y. Shi, P. Zhong, **P. Xie**[†], “Intrinsic structure of relaxor ferroelectrics from first principles”, Phys. Rev. B
2. **P. Xie**, Y. Chen, X. Xu, Z. Yao, W. E and R. Car, “Ab Initio bulk free energy surface of proper ferroelectrics” Phys. Rev. Lett. 136.5 (2026): 056801 [**EDITORS’ SUGGESTION**].
3. **P. Xie**, K. Wang, A. Mitra, Y. Zhu, X. Li, W. A. de Jong and C. Yang, “Predicting open quantum dynamics with data-informed quantum-classical dynamics” Phys. Rev. Lett. 136.1 (2026): 010402.

Doctoral Research

4. **P. Xie**, Y. Chen, W. E and R. Car. “Thermal disorder and phonon softening in the ferroelectric phase transition of lead titanate” Phys. Rev. B 111.9 (2025): 094113 [**EDITORS’ SUGGESTION**].
5. B. Yang, **P. Xie**, and R. Car. “Deuteration removes quantum dipolar defects from KDP crystals.” npj Comput. Mater. 10.1 (2024): 241
6. **P. Xie** and W. E. “Coarse-graining conformational dynamics with multi-dimensional generalized Langevin equation: how, when, and why.” J. Chem. Theory Comput. 20.18 (2024): 7708–7715
7. **P. Xie**, R. Car, and W. E. “Ab Initio Generalized Langevin Equations .” Proc. Natl. Acad. Sci. 121.14 (2024): e2308668121
8. W. E, H. Lei, **P. Xie**, L. Zhang “Machine Learning-Assisted Multi-scale Modeling .” J. Math. Phys. 64.7 (2023): 071101
9. **P. Xie**, and W. E. “Coarse-grained spectral projection: A deep learning assisted approach to quantum unitary dynamics.” Phys. Rev. B 103.2 (2021): 024304.
10. R. Wu, X. Cao, **P. Xie**, and Y. Liu. “End-to-end quantum machine learning implemented with controlled quantum dynamics.” Phys. Rev. Appl. 14.6 (2020): 064020.

Undergraduate Research

11. **P. Xie**, B. Yang, Z. Zhang and R. F. S. Andrade. “Exact evaluation of the causal spectrum and localization properties of electronic states on a scale-free network ” Physica A 502 (2018): 40-48.
12. B. Yang, **P. Xie** and Z. Zhang. “Effects of heterogeneity in site-site couplings for tight-binding models on scale-invariant structures.” Phys. Lett. A 381.44 (2017): 3773-3778.
13. **P. Xie**, B. Wu and Z. Zhang. “Exactly solvable tight-binding model on two scale-free networks with identical degree distribution.” EPL 116.3 (2016): 38002.
14. **P. Xie**, Z. Zhang and F. Comellas. “The normalized Laplacian spectrum of subdivisions of a graph.” Appl. Math. Comput. 286 (2016): 250-256.
15. **P. Xie**, Z. Zhang and F. Comellas. “On the spectrum of the normalized Laplacian of iterated triangulations of graphs.” Appl. Math. Comput. 273 (2016): 1123-1129.

16. **P. Xie**, Y. Lin and Z. Zhang. "Spectrum of walk matrix for Koch network and its application." J. Chem. Phys. 142.22 (2015): 224106.

CONFERENCE PRESENTATIONS

1. (Featured & Awarded) International Workshop on Relaxor Ferroelectrics, Himeji, May 2026
2. MRS Annual Meeting, Honolulu, April 2026
3. APS March Meeting, Denver, March 2026
4. (Invited) Workshop on Mathematics and AI, National University of Singapore, February 2026
5. (Invited) Lennard-Jones Centre-CECAM Meeting, University of Cambridge, September 2025
6. APS March Meeting, Anaheim, March 2025
7. Annual Meeting of the Biophysical Society, Los Angeles, February 2025
8. 36th Fundamental Physics of Ferroelectrics Workshop, Annapolis, February 2025
9. Simons Symposium: Challenges in Biomolecular Simulations, NYU, New York, May 2024
10. APS March Meeting, Minneapolis, March 2024
11. 35th Fundamental Physics of Ferroelectrics Workshop, Washington, D.C., February 2024
12. International Workshop on Nuclear Quantum Effects, NYU, New York, June 2023
13. APS March Meeting, Las Vegas, March 2023
14. APS March Meeting, Chicago, March 2022
15. 33rd Fundamental Physics of Ferroelectrics Workshop, Washington, D.C., February 2022
16. U21 Undergraduate Research Conference, University of Edinburgh, Edinburgh, June 2017

INVITED SEMINAR TALKS

1. GM on Zoo Seminar, UChicago, April 2026
2. Quantum 290 Seminar, UC Berkeley, February 2026
3. Physical Chemistry Seminar, UC Irvine, May 2025
4. BIDMaP Seminar, UC Berkeley, May 2025
5. National University of Singapore, April 2025 (Virtual)
6. Ruhr-Universität Bochum, March 2025 (Virtual)
7. Applied Math Seminar, Stanford University, February 2025
8. Pitzer Center Theoretical Chemistry Seminar, UC Berkeley, October 2024
9. Computing Science Seminar, Lawrence Berkeley National Lab, January 2024
10. Chemistry in Solution and at Interfaces Seminar, Princeton University, March 2023
11. Samsung Semiconductor Advanced Materials Lab, Boston, August 2022
12. PACM Colloquium, Princeton University, February 2022
13. Chemistry in Solution and at Interfaces Seminar, Princeton University, February 2022